INCH-POUND

MIL-S-51235C
8 June 1989
SUPERSEDING
MIL-S-51235B
26 January 1988

MILITARY SPECIFICATION

SMOKE POT, FLOATING, SGF2, AN-M7A1

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of smoke pot in two conditions.

1.2 Classification. The smoke pot shall be in one of the following conditions as specified (see 6.2):

Condition A - With fog oil, fuel block assembly, and fuze
Condition B - Without fog oil, fuel block assembly, and fuze

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Chemical Research, Development and Engineering Center, ATTN: SMCCR-SPT-S, Aberdeen Proving Ground, MD 21010-5423 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1365

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SPECIFICATIONS

MILITARY

MIL-F-11522 - Fuze, Smoke Pot, M208 and Fuze, Hand Grenade, M226
MIL-F-11673 - Fuze, Smoke Pot, Floating, Electric, M209
MIL-F-12070 - Fog Oil

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

U.S. ARMY CHEMICAL RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

DRAWINGS

36-2-212 - Container (Metal Parts Assembly)
36-2-214 - Smoke Pot, Floating, SGF2, AN-M7A1

(Copies are available from Commander, U.S. Army Information Systems Command, ATTN: ASNC-ARI-IMC-TM, Rock Island, IL 61299-7300.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials and components.

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3.1.1 Materials. All materials cited on Drawing 36-2-214 and on the subsidiary drawings shall conform to the specifications listed thereon or to the specific characteristics set forth on the drawings.

3.1.2 Components. All components of the smoke pot shall conform to the specifications and drawings listed on Drawing 36-2-214 and on subsidiary drawings.

3.2 Assembly (see 6.3).

3.2.1 Condition A. The smoke pot shall be assembled and filled as shown on Drawing 36-2-214 using fog oil, type SGF2 conforming to MIL-F-12070.

3.2.2 Condition B. The smoke pot shall be assembled as shown on Drawing 36-2-212.

3.3 Leakage. The smoke pot container (Drawing 36-2-212) with the fuel block assembly (Drawing 36-2-219) installed but less the fog oil, outer cover (Drawing 36-2-195) and cover lug (Drawing 36-2-196) shall have a pressure drop rate no greater than 1 inch of water per minute at a pressure differential of 10.0 ± 0.5 inches of mercury when tested as specified in 4.2.4.1.

3.4 Functioning. The smoke pot shall ignite and emit a dense cloud of smoke within a maximum period of 50 seconds. The pot shall continue to emit under pressure a dense cloud of smoke for a period of no less than 8 nor more than 13 minutes without any interruption greater than 80 seconds, from the time the fuel block begins to burn until the smoke is no longer emitted under pressure when tested as specified in 4.2.4.2.

3.5 Workmanship. The smoke pot shall be free from foreign matter (oil or grease) and damage such as dents, cracks, or deformation which could affect its use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection require-

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ments in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Objective evidence. The contractor shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the smoke pots of one interfix number, produced by one manufacturer, at one plant, from the same materials and under essentially the same manufacturing conditions. However, no more than one lot of fuzes nor more than one lot of each sustainer or propellant mix ingredients, produced by one manufacturer shall be represented in any one lot of assembled smoke pots.

4.2.2 Sampling.

4.2.2.1 For examination and nondestructive tests. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.2.2 For functioning tests. Sampling shall be conducted in accordance with table I.

TABLE I. **Acceptance/rejection criteria**

Lot size	To 1200	1201 to 3200
Sample size	32	50
Test paragraph	Accept/reject No.	Accept/reject No.
4.2.4.2(a)	1 2	1 2
4.2.4.2(b)	1 2	1 2
4.2.4.2(c)	1 2	1 2
4.2.4.2(d)	1 2	1 2

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4.2.3 Inspection.

4.2.3.1. For examination and nondestructive tests. Sample smoke pots and components shall be examined and tested in accordance with the classification of defects and MIL-STD-105.

4.2.3.2 For functioning tests. The whole sample of smoke pots shall be tested in accordance with 4.2.4.2 including all the characteristics listed in table I (see 6.3).

4.2.3.3 Critical defects. Each smoke pot in the lot shall be inspected for the critical characteristics listed in the classification of defects.

4.2.3.4 Classification of defects.

(a) Smoke pot, floating, SGF2, AN-M7A1 (Drawing 36-2-214)

Categories	Defects
Critical:	
1	Fuze safety pin not spread (applicable to M208 fuze only)
Major:	AQL 1.0 percent defective
101	Component missing
102	Component incorrectly assembled
103	Marking incorrect or missing
104	Instruction card missing
105	Workmanship (3.5)

(b) Container (Drawing 36-2-212)

Categories	Defects	Acceptance standards
Critical:	None defined	
Major:	100 percent inspection	
101	Leakage	4.2.4.1
Major:	AQL 1.0 percent defective	
102	Component missing	

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103	Component incorrectly assembled
104	Fuze adapter thread incorrect
105	Marking missing or incorrect
106	Cap-plug missing
107	Workmanship (3.5)

(c) Fuel block assembly (Drawing 36-2-219)

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Component incorrectly assembled
102	Top surface of fuel block not coated
103	0.75 diameter hole missing
104	Height of container incorrect
105	Diameter (7.85) incorrect
106	Workmanship (3.5)

4.2.4 Tests.

4.2.4.1 Leakage. The smoke pot as described in 3.3 shall be evacuated through both the oil filler opening and the venturi to the specified pressure differential. The vacuum source shall be shut off at the venturi (gage must be between the shut off valve and the venturi) and the oil filler opening (gage must be between the shut off valve and the oil filler opening). The vacuum shall be maintained for a minimum period of 1 minute. The pressure drop shall be measured and recorded. A pressure loss greater than specified shall be cause for rejection. An independent leakage test shall be performed on each chamber.

4.2.4.2 Functioning. Sample smoke pots or containers filled with SGF2 fog oil conforming to MIL-F-12070 shall be fuzed with M208 fuzes conforming to MIL-F-11522 or M209 fuzes conforming to MIL-F-11673 and functioned in water. The depth of the water shall be a minimum of 2 feet. The functioning shall be observed for defects as follows (see 6.3):

(a) Failure to ignite or excessive interruption of dense cloud of smoke
(b) Excessive delay in emission of dense cloud of smoke
(c) Duration of smoke emission less than the minimum specified
(d) Duration of smoke emission greater than the maximum specified

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5. PACKAGING

5.1 Palletization. The level of palletization shall be as specified (see 6.2).

5.1.1 Level A. Smoke pot assemblies shall be palletized in accordance with Drawing 36-1-119.

5.1.2 Level C. Smoke pot assemblies shall be palletized in accordance with Drawing 36-1-119 except pallets shall not be treated with wood-preservative.

5.2 Marking. In conjunction with information furnished in the contract or order, marking shall be as specified in 5.2.1 and 5.2.2.

5.2.1 Smoke pots. Smoke pots shall be marked in accordance with Drawing 36-2-214 as applicable to the fuzing and condition type.

5.2.2 Palletized loads. Palletized loads shall be marked in accordance with Drawing 36-1-119 as applicable to the fuzing and condition type.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The smoke pot is intended for use in producing smoke screens on land or on water. It can be fired manually or by remote control electric current.

6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of this specification
(b) Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1)
(c) Condition of smoke pot required (see 1.2)
(d) If M208 or M209 fuze is required
(e) Additional instruction cards to be furnished for the Navy
(f) Level of palletization required (see 5.1)

6.3 Caution. Due to the hazard involved in filling items with flammable mixtures, contractors are advised to exercise all necessary precautions to protect personnel from injury during filling and testing operations.

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6.4 Components specifications. The following specifications reflect the component required for the complete assembly of the smoke pot:

MIL-F-11522 - Fuze, Floating Smoke Pot, M208 and Fuze, Hand Grenade, M226
MIL-F-11673 - Fuze, Smoke Pot, Floating, Electric, M209

6.5 Subject term (key word) listing.

Fog oil
Fuel block assembly
Fuze, M208
Fuze, M209

Custodians:

Army - EA
Navy - OS

Preparing activity:

Army - EA

Project No. 1365-0244

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-51235C	SMOKE POT, FLOATING, SGF2, AN-M7A1

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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